

# Use Case Descriptions: UC01 to UC05

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## UC01: Patient Admission from OPD to Ward

| Field         | Detail                                                                                                                            |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID   | UC01                                                                                                                              |
| Use Case Name | Patient Admission from OPD to Ward                                                                                                |
| Actors        | OPD Doctor (Primary), Ward Doctor, Nurse, System                                                                                  |
| Related FRs   | FR01, FR02, FR09                                                                                                                  |
| Survey Basis  | 92% of respondents reported no centralised patient history; 100% experienced delayed diagnosis linked to poor OPD toward handoff. |

### Preconditions

- Patient has been assessed at OPD.
- OPD Doctor is authenticated in the system.
- At least one ward bed is available (see UC02).
- Design Constraint DC01 applies: System must accept BHT Number as a secondary identifier for backward compatibility with existing paper Health 26 forms.

### Main Flow

1. OPD Doctor selects "Admit Patient" from the patient's OPD encounter record.
2. System displays the patient's existing medical history (if any) and current OPD observations.
3. OPD Doctor enters admission data, mapping to Health 26 Section A and B fields:  
Ward/Unit and Consultant Name (Section A)  
Mode of Admission: Direct / Transferin / Requested (Section B)  
Provisional Diagnosis with ICD10/11 code lookup (Section B) (DC07)  
Level of Consciousness, Pulse Rate, Blood Pressure (Section B)
4. System queries available wards and bed types matching the required care level.
5. OPD Doctor selects a ward; system triggers bed allocation (UC02).
6. System generates an admission record linked to the patient's persistent medical history (UC09) and assigns a BHT Number for this admission (DC01).
7. System notifies the assigned ward doctor and duty nurse of the incoming patient (UC07).
8. OPD Doctor confirms admission; system timestamps and locks the OPD encounter.
9. Patient status changes to "Admitted In Transit."

## Alternative Flows

### AF01A: No Beds Available

- At step 4, if no beds match the required care level:
  - System displays "No beds available" with a waitlist option.
  - OPD Doctor may place the patient on a priority waitlist.
  - System notifies the ward nurse when a bed becomes available.
  - Resume at step 5 once bed is confirmed.

### AF01B: Patient Requires Emergency Escalation

- At step 3, if the OPD Doctor flags the case as an emergency:
  - System bypasses normal waitlist and reserves the next available ICU or highdependency bed.
  - System immediately alerts the oncall ward doctor via UC07.
  - Admission is marked "Emergency Priority."

### AF01C: Returning Patient with Existing Record

- At step 2, if the system detects a prior admission:
  - System surfaces previous diagnoses, allergies (from Section E), and medication history.
  - OPD Doctor reviews and confirms carryforward information before proceeding to step 3.

## Postconditions

- Patient admission record is created and linked to the patient's full medical history.
- A BHT Number is assigned for this admission.
- Ward doctor and duty nurse are notified.
- Bed is reserved and removed from the available pool.

## UC02: Bed Allocation

| Field         | Detail                               |
|---------------|--------------------------------------|
| Use Case ID   | UC02                                 |
| Use Case Name | Bed Allocation                       |
| Actors        | Nurse (Primary), Ward Doctor, System |
| Related FRs   | FR02                                 |

### Preconditions

- An admission request has been initiated (UC01 or emergency workflow).
- The nurse is authenticated and assigned to the relevant ward.

### Main Flow

10. System receives a bed allocation request from UC01 or directly from a nurse.
11. System displays a realtime ward map showing bed status: Available / Occupied / Reserved / Under Maintenance.
12. Nurse selects the most appropriate bed based on patient needs (gender, care level, isolation requirement).
13. Nurse confirms bed selection.
14. System updates bed status to "Reserved" and associates it with the incoming patient.
15. System records the nurse's ID, timestamp, and bed selection in the audit log.
16. System notifies the OPD admitting doctor that a bed has been confirmed.
17. When the patient physically arrives at the ward, the nurse updates the status from "Reserved" to "Occupied."

### Alternative Flows

#### AF02A: Nurse Overrides System Suggestion

- At step 3, if the nurse selects a bed different from the system's suggested optimal bed:
  - System prompts for a reason (e.g., patient preference, equipment proximity).
  - Nurse enters reason; system logs the override with the nurse's credentials.
  - Allocation proceeds normally.

#### AF02B: Bed Requires Maintenance Before Use

- At step 3, if the nurse flags the selected bed as requiring maintenance:
  - System marks the bed "Under Maintenance Pending" and raises a maintenance notification.
  - Nurse selects an alternative bed; allocation resumes at step 4.

#### AF02C: Ward at Full Capacity

- At step 2, if all beds in the target ward are occupied or reserved:
  - System displays a capacity alert and suggests wards with available beds of equivalent care level.

- If no ward can accommodate, the system escalates to the Ward Manager role.

### Postconditions

- Bed is associated with the patient and removed from the available pool.
- Ward capacity statistics are updated in real time.
- Audit trail records who allocated the bed and when.

## UC03: Doctor Observation and Treatment Update

| Field         | Detail                                  |
|---------------|-----------------------------------------|
| Use Case ID   | UC03                                    |
| Use Case Name | Doctor Observation and Treatment Update |
| Actors        | Ward Doctor (Primary), Nurse, System    |
| Related FRs   | FR03, FR04, FR07                        |

### Preconditions

- Patient is admitted and assigned to the ward doctor's care list.
- Ward doctor is authenticated.
- Patient admission record exists (UC01).

### Main Flow

18. Ward Doctor opens the patient's current record from their worklist.
19. System displays the patient's admission notes, prior observations, active medications, and any pending lab results.
20. Ward Doctor enters new clinical observations, mapping to Health 26 Section C fields (DC02):  
Level of Consciousness: Alert / Responds to Voice / Responds to Pain / Unconscious  
Temperature (°F/°C), Pulse Rate (/min), Respiratory Rate (/min)  
Blood Pressure (mmHg), SpO<sub>2</sub> (%), CBS (mg/dl)  
Weight (kg), Height (cm), BMI (autocalculated by system)  
Urine WD test result
21. Ward Doctor updates or adds treatment orders (medications, procedures, referrals).
22. Ward Doctor enters a followup instruction (e.g., "Review in 4 hours", "Await lab results").
23. System timestamps and saves the update, appending it to the patient's longitudinal record.
24. System notifies the duty nurse of any new medication or procedure orders (UC07).
25. If a lab test is ordered, system creates a lab request (UC05).

### Alternative Flows

#### AF03A: Critical Observation Requiring Immediate Escalation

- At step 3, if the Ward Doctor marks an observation as "Critical":
  - System immediately alerts the oncall senior doctor via UC07 (P1 notification).
  - An escalation flag is added to the patient's record.
  - The normal update flow continues in parallel.

#### AF03B: Conflicting Medication Order Detected

- At step 4, if the system detects a conflict with an existing medication (allergy flag from Section E per DC04, or drug interaction):
  - System displays a clinical alert with severity level and conflict description.

- Ward Doctor must explicitly acknowledge and override, or modify the order.
- Override is logged with the doctor's ID and stated reason.

#### **AF03C: Offline Mode**

- If network connectivity is lost during the update:
  - System caches the observation locally and displays a syncpending indicator.
  - Data is synchronised automatically when connectivity is restored.
  - Record is flagged "Pending Sync" to prevent concurrent edits.

#### **Postconditions**

- Observation and treatment update are appended to the patient's longitudinal record.
- Nursing staff are notified of any actionable orders.
- Any lab test orders are forwarded to the laboratory (UC05).

## UC04: Nurse Medication Tracking (Tick System)

| Field         | Detail                                  |
|---------------|-----------------------------------------|
| Use Case ID   | UC04                                    |
| Use Case Name | Nurse Medication Tracking (Tick System) |
| Actors        | Nurse (Primary), Ward Doctor, System    |
| Related FRs   | FR04                                    |

### Preconditions

- A medication order has been entered by a Ward Doctor (UC03).
- Nurse is authenticated and on duty in the relevant ward.

### Main Flow

26. Nurse opens the Medication Administration Record (MAR) for their assigned patients.
27. System displays a list of due medications for each patient, sorted by scheduled time.
28. MAR columns mirror the observed paper sheet (DC05): Drug Name, Dosage, Route, Frequency with digital additions: Scheduled Time, Nurse ID (autopopulated from login), Timestamp, and Status.
29. Nurse selects a medication row for a specific patient.
30. System displays: drug name, dose, route, scheduled time, prescribing doctor, and any allergy warnings sourced from Section E (DC04).
31. Nurse administers the medication to the patient.
32. Nurse records administration with a single tap: selects "Administered" and confirms. UX requirement: administration must be a singletap action matching the simplicity of the paper tick. It must not require multiple screens.
33. System timestamps the administration, records the nurse's ID, and marks the row green.
34. System removes the medication from the "Due Now" list.
35. If the medication is the final dose in a prescribed course, system flags the record for doctor review.

### Alternative Flows

#### AF04A: Medication Withheld

- At step 6, if the nurse cannot administer (patient refused, patient unavailable, clinical contraindication):
  - Nurse selects "Withheld" and enters a mandatory reason.
  - System marks the row amber and notifies the prescribing ward doctor.
  - The missed dose is logged for the doctor's next review cycle.

#### AF04B: Partial Dose Administered

- At step 6, if only a partial dose can be given:
  - Nurse selects "Partial" and enters the actual dose administered.
  - System logs the deviation and alerts the prescribing doctor.

- The remaining dose is rescheduled per doctor instruction.

#### **AF04C: Overdue Medication Alert**

- If a medication passes its scheduled time without a tick:
  - System escalates the alert: yellow at 15 minutes overdue, red at 30 minutes overdue.
  - Ward charge nurse receives a notification at 15 minutes.
  - If unresolved at 45 minutes, the ward doctor is notified.

#### **Postconditions**

- Full medication administration record is maintained with timestamps and nurse IDs.
- Any deviations (withheld, partial) are visible to the prescribing doctor.
- Audit trail is immutable and available for clinical governance review.



## UC05: Lab Test Request

| Field         | Detail                                               |
|---------------|------------------------------------------------------|
| Use Case ID   | UC05                                                 |
| Use Case Name | Lab Test Request                                     |
| Actors        | Ward Doctor (Primary), OPD Doctor, Lab Staff, System |
| Related FRs   | FR05, FR06                                           |

### Preconditions

- Patient is registered in the system (admitted or OPD visit in progress).
- Requesting doctor is authenticated.
- Laboratory department is operational and reachable on the network.

### Main Flow

36. Doctor opens the patient's record and selects "Request Lab Test."
37. System displays a catalogue of available tests grouped by category (Haematology, Biochemistry, Microbiology, etc.).
38. Doctor selects one or more tests and sets urgency level: Routine / Urgent / STAT.
39. System autopopulates patient demographics, requesting doctor identity, and timestamp.
40. Doctor confirms and submits the request.
41. System creates a lab request record and assigns a unique Request ID linked to the patient's persistent ID and current BHT Number (DC01).
42. System sends the request to the laboratory worklist in real time (UC06).
43. Doctor receives a confirmation: "Lab request [ID] submitted [Test Name] [Urgency]."

### Alternative Flows

#### AF05A: STAT Request Immediate Escalation

- At step 3, if the doctor selects urgency "STAT":
  - System highlights the request in red on the lab worklist.
  - System sends an immediate push notification to the on duty lab technician (UC07).
  - System activates a 30minute SLA countdown timer for the STAT request.

#### AF05B: Test Not in Catalogue

- At step 2, if the required test is not listed:
  - Doctor selects "Request Unlisted Test" and enters a freetext description.
  - System routes the request to the lab supervisor for review before adding to the worklist.
  - Doctor is notified once the lab supervisor accepts or rejects the custom request.

#### AF05C: Duplicate Request Detection

- At step 5, if an identical test for the same patient was requested within the past 24 hours:

- System warns: "Duplicate detected [Test Name] requested at [Time] by [Doctor]."
- Doctor may proceed, cancel, or view the result of the prior request.

### Postconditions

- Lab request is created with a unique ID, linked to the patient's persistent record and BHT Number.
- Requesting doctor receives confirmation.
- Lab staff receive a realtime notification and the request enters the lab workflow (UC06).